

QuantumAegis: Execution Brief

The Problem

Blockchain infrastructure secures \$3T+ in assets using cryptography that quantum computers will break. RSA, ECDSA, BLS—all vulnerable to Shor's algorithm. Timeline: 10-15 years to cryptographically relevant quantum computers. The "harvest now, decrypt later" attack is already active. Every on-chain transaction is a permanent liability.

Adjacent problem: AI systems make financial decisions without cryptographic accountability. No way to verify a model produced a claimed output without exposing proprietary weights. Trust model incompatible with decentralized infrastructure.

The Market

Segment	2025	2035	CAGR
Post-Quantum Cryptography	\$0.42B	\$13.3B	46%
Blockchain Security	\$3.0B	\$23.4B	38%
Quantum Cybersecurity	\$2.0B	\$24.2B	32%
Verifiable AI (zkML)	\$0.1B	\$8.0B	55%
Combined TAM	\$5.5B	\$68.9B	~35%

What We Build

Layer 1: Quantum Coprocessor

- Off-chain quantum circuit simulation (Google Cirq, Willow 105Q model)
- Threat assessment engine: Grover (symmetric) + Shor (asymmetric) attack modeling
- Risk scoring across 12 vulnerability categories
- Proof generation: Poseidon2 hashes + ML-DSA-87 signatures

Layer 2: Post-Quantum Cryptography

- ML-DSA-87 (Dilithium-5): Lattice-based signatures, NIST Level 5
- SLH-DSA-256s (SPHINCS+): Hash-based signatures, stateless
- Hybrid ECDSA+PQC: Backward compatibility during transition
- Automated algorithm rotation triggered by threat thresholds

Layer 3: TEE Sequencer

- Aegis-TEE: Intel TDX/SGX, AMD SEV enclave execution
- Encrypted mempool: Transactions hidden until ordering committed
- Hardware attestation chain verified on-chain
- Phala Network redundancy for failover

Layer 4: On-Chain Verification

- OP Stack L2 rollup (Chain ID 16584, Sepolia L1)
- PQCVerifier contract: ~48,000 gas per verification
- QRMSOracle: On-chain threat indicator registry

- Result immutability + audit trail

How We Ship

Phase	Deliverable	Status
Foundation	OP Stack L2, QRMS service, contracts, dashboard	Complete
Cryptography	ML-DSA-87, SLH-DSA-256s, hybrid signatures	Complete
TEE Integration	Aegis-TEE sequencer, Phala redundancy	Complete
QVM Oracle	Cirq simulation, Grover/Shor assessment	Complete
Threat Intel	12 categories, risk scoring, feeds	Simulated
Quantum Coprocessing	Off-chain compute, on-chain verify	In Progress
zkML Integration	Verifiable AI inference proofs	Planned
Production	Mainnet, audits, monitoring	Q1 2027

Competitive Position

No existing L2 integrates quantum threat assessment with adaptive PQC and TEE sequencing. Competitors address fragments: QRL provides quantum-resistant L1 (no EVM compatibility), Phala offers TEE compute (no PQC), zkSync/Optimism provide rollups (classical crypto only). QuantumAegis is the composable security layer—works alongside OP Stack's 10-year ECDSA deprecation roadmap, integrates with Lean Ethereum (leanVM, leanSig), provides direct-to-consumer threat APIs.

Defensibility

- **Integration depth:** Four-layer stack creates switching costs. Each layer reinforces the others.
- **Data flywheel:** Threat assessments improve with usage. First-mover captures training data for ML risk prediction.
- **Standards alignment:** NIST PQC, OP Stack roadmap, EIP-7702—we build on what's shipping, not theoretical.
- **Time asymmetry:** Migration takes years; we're shipping now. Late entrants face compressed timelines.

Execution Requirements

Engineering: Rust (coprocessor), Solidity (contracts), Go (OP Stack mods). **Infrastructure:** TEE-capable cloud (Azure Confidential, GCP Confidential VMs), quantum simulation capacity. **Integrations:** Wallet providers, bridge operators, DeFi protocols for adoption. **Compliance:** Security audit (Trail of Bits tier), regulatory engagement for PQC migration guidance.

Bottom Line

\$69B TAM by 2035. Quantum breaks current crypto. Migration takes years. We have working infrastructure—L2 operational, PQC integrated, TEE sequencer complete, QVM oracle functional. Gap to close: production hardening, zkML integration, mainnet deployment. The window for proactive migration is finite. Ship now or scramble later.